Derivable Judgement

Statistical Decision-Making Model — Theory Report (Part A)

1. What is Inferential Statistics?

Inferential statistics draws conclusions about a population from a sample. It uses estimation (point estimates and confidence intervals) and hypothesis testing, and always quantifies the uncertainty of the conclusion.

2. Hypothesis Testing and its Components

Hypothesis testing is a formal procedure to decide between two competing claims using sample data. Components: **Null hypothesis (H_0)** — no effect/difference; **Alternative hypothesis (H_1)** — an effect/difference exists; **Significance level (α)** — tolerated probability of a false rejection (commonly 0.05); **Test statistic** — value computed from the sample; **Critical value / p-value** — basis for the decision; **Decision** — reject or fail to reject H_0.

3. Confidence Interval and Critical Value

A confidence interval is a range of values likely to contain the true population parameter at a stated confidence level (e.g. 95%). The **critical value** is the cut-off point on the test distribution (z, t, χ^2 or F) that marks the rejection region; if the test statistic falls beyond it, H_0 is rejected.

Formula: $CI = \bar{x} \pm (\text{critical value}) \times (s / \sqrt{n})$

4. p-value

The p-value is the probability of obtaining a result at least as extreme as the one observed, assuming H_0 is true. A small p-value ($p < \alpha$) is evidence against H_0, so H_0 is rejected; a large p-value means we fail to reject it.

5. Type I vs Type II Errors

Type I error (α) — rejecting H_0 when it is actually true (false positive). **Type II error (β)** — failing to reject H_0 when it is actually false (false negative). Reducing one generally increases the other; power of a test = $1 - \beta$.

6a. z-test

Used to compare a sample mean/proportion with a population value, or two means, when the sample is large ($n \geq 30$) or the population standard deviation is known. Statistic follows the standard normal distribution.

Formula: $z = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$

6b. t-test

Used to compare means when the sample is small or the population standard deviation is unknown. Types: one-sample, independent two-sample (Student/Welch) and paired. Statistic follows the t-distribution with the appropriate degrees of freedom.

Formula: $t = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$

6c. Chi-square test

Tests whether two categorical variables are associated (test of independence) or whether observed frequencies fit an expected distribution (goodness-of-fit), by comparing observed and expected counts.

Formula: $\chi^2 = \Sigma\ (O - E)^2 / E$

6d. ANOVA

Analysis of Variance compares the means of three or more groups simultaneously, avoiding the inflated Type I error of many t-tests. It uses the F-statistic: ratio of between-group variance to within-group variance.

Formula: $F = MS(between) / MS(within)$

7. Covariance

Covariance measures how two variables change together. Positive covariance means they tend to rise together; negative means one rises as the other falls. Its magnitude depends on the units, so it is hard to compare across variable pairs.

Formula: $Cov(X, Y) = \Sigma\ (x_i - \bar{x})(y_i - \bar{y}) / (n - 1)$

8. Correlation

Correlation (Pearson r) is the standardised covariance. It ranges from −1 (perfect negative) to +1 (perfect positive linear relationship); 0 means no linear relationship. Correlation does not imply causation.

Formula: $r = Cov(X, Y) / (s_x \cdot s_y)$

Summary of Test Selection

Question	Variables	Test
Do two group means differ (large n)?	Numeric vs 2 groups	z-test
Do two group means differ (small n)?	Numeric vs 2 groups	t-test
Are two categories associated?	Categorical vs categorical	Chi-square
Do 3+ group means differ?	Numeric vs 3+ groups	ANOVA
Are two numeric variables related?	Numeric vs numeric	Covariance / Correlation